

### 3.4 BANDWIDTH CONTROLS OPTIMIZATION ACCESS TO HIGH PRECISION

The construction’s nearly width-independent relative bandwidth  $\lambda = \gamma h$  requires  $\gamma = \Theta(W)$ . Even though larger widths admit smaller  $\lambda$  at comparable construction error (Figure 2b), small slopes can delay readout optimization. We quantify this in the dictionary of Section 3.1, replacing the direct normalized least-squares solve with GD while keeping the features fixed.

**Theorem 3.2** (Slope-dependent output-error lower bound). *For the fixed-center construction of Section 3.1 with common slope  $\gamma > 0$  and nonzero sampled target, readout GD from zero with  $\eta = 1/(2\mu_1)$  satisfies*

$$\frac{\|f - q_n\|_{2,Z}^2}{\|f\|_{2,Z}^2} \geq \sum_i p_i(\gamma) \left(1 - \frac{\bar{\rho}_i(\gamma)}{2}\right)^{2n}, \quad n \geq 0. \quad (1)$$

Here  $\|\cdot\|_{2,Z}$  is the training-sample RMS norm,  $\mu_i$  are the descending readout-kernel eigenvalues,  $p_i$  is the fraction of target energy in eigenvector  $i$ , and  $\bar{\rho}_i(\gamma) \in [0, 1]$  bounds  $\mu_i/\mu_1$  from above. The explicit gamma-dependent rate bounds and finite-geometry allowances are given in Appendix A.

**Proof sketch.** Gamma enters the feature correlations through  $d \coth(\gamma d)$ , where  $d$  is the separation between training points. Controlling the finite-center corrections gives the relative-rate bounds; a standard spectral argument then yields the output-error inequality.

Increasing  $\lambda$  from  $1/32$  to  $1/4$  raises the thirtieth relative eigenvalue by  $1.4 \times 10^5$ , versus only 1.3 for the fourth (Figure 3); the thirtieth mode carries about 2.5% of target energy at both endpoints. At  $\lambda = 1/32$ , a direct readout fit attains  $6.2 \times 10^{-10}$  dense-grid relative error, yet five million GD updates retain training error 0.210 against the theorem’s lower bound 0.2098 (Figure 4A). At  $\lambda = 1/4$ , GD reaches  $3.88 \times 10^{-4}$  over the same budget. Across the four bandwidths, the bound remains at least 75% of executed error throughout training: small slopes delay accuracy already available in the dictionary.

Over the same budget, joint Adam reaches median dense-grid relative error  $1.81 \times 10^{-3}$ , versus  $6.62 \times 10^{-7}$  on the supplied  $\lambda = 1/4$  features, despite partial slope acquisition (Figure 4B–C). This motivates the next analysis: training must acquire useful slopes and centers together. The construction supplies both the feature geometry and the readout directly.